

ECON 494 Spatial Data Science

Conflict Dynamics in the Sahel Region

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Description of the Project

Our analysis aims at understanding the current dynamics of the ongoing conflicts in the Sahel region, especially in Mali, where armed groups (JNIM and ALF) are gaining terrain against the official government. Our objectives were to overview the repartition of violences committed against civilians, both geographically and in terms of attributability.

- The first part of our work focuses on the Mali case, and researches for periodicity and location of these violent phenomena.
- The second part expands the analysis on political violence in the Sahel region, and shows the repartition by geographic area, along with conflict intensity and severity.
- The third part aims at understanding if there is a link between the presence of natural resources or urban area in the region and the repartition of political violence.

Nature of Data

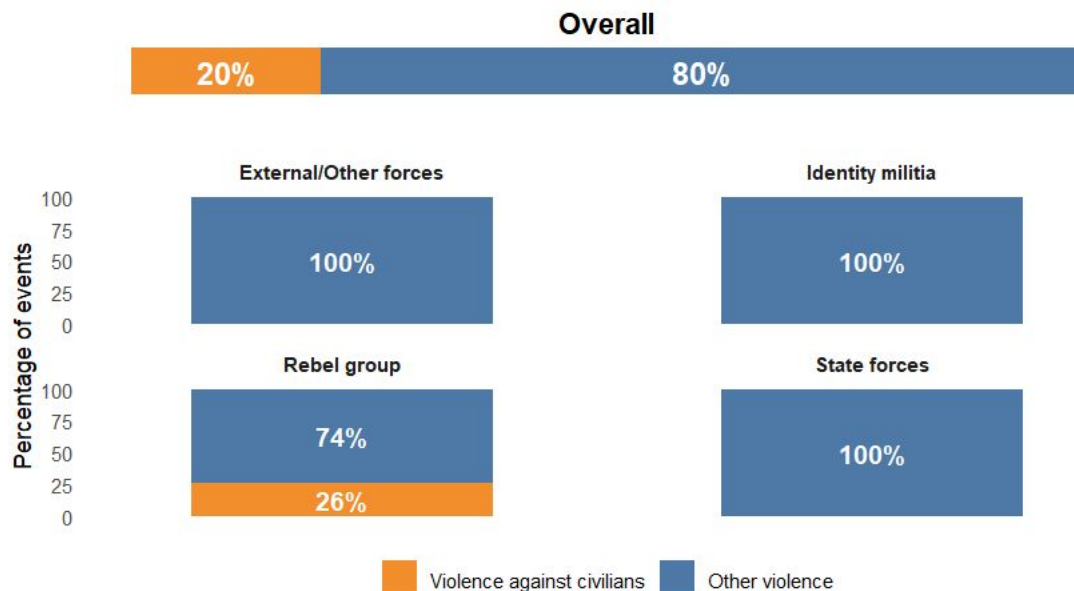
We downloaded the ACLED dataset for 2025, as it is the most up-to-date

Variables used:

1. Mali case
 - identify violence type: Violence against civilians; Other violence
 - identify actors: External/Other forces; Rebel group; Identity militia; State forces
 - identify number of events and fatalities
2. Sahel region
 - identify the extended event type: Battles; Explosions/Remote violence; Violence against civilians
 - identify 2 areas: Central Sahel; Southern frontier
3. Relationship with factors
 - identify mining areas
 - identify distance to cities

Percentage of organized political violence events targeting civilians by actor

Percentage of organized political violence events against civilians by actor:
1 January 2025 – 28 April 2025

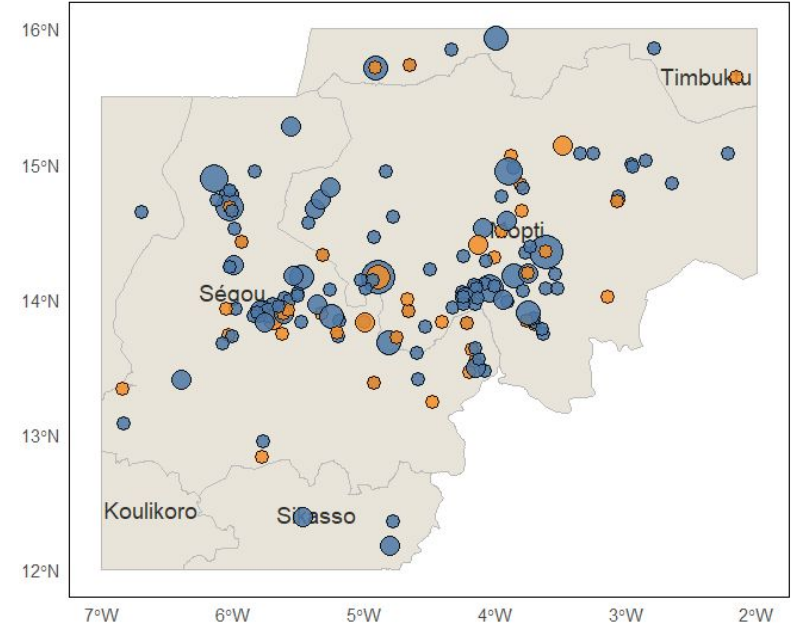
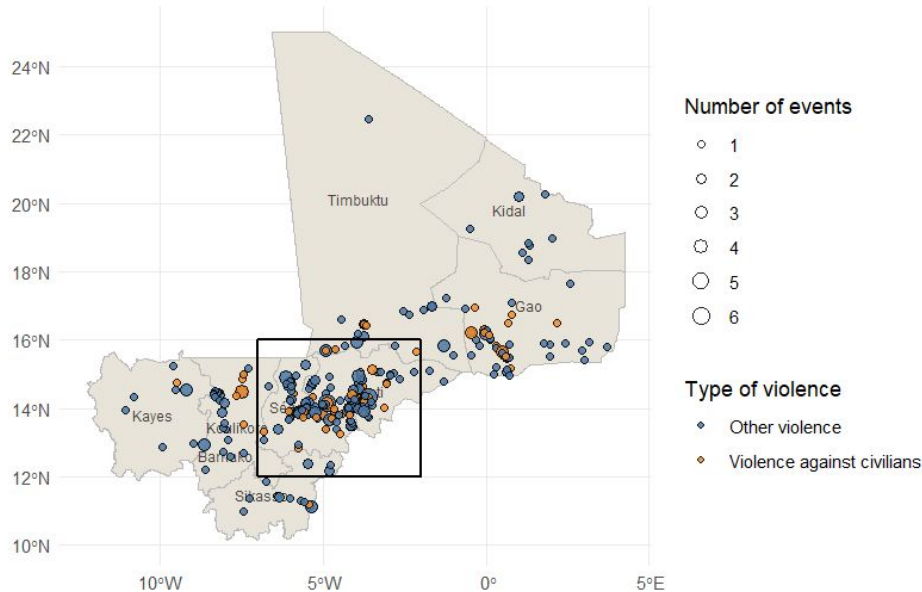


With Wagner's departure in 2025

- Only the Rebel group shows violence against civilians
- State violence against civilians has stopped
- Two new groups also do not show violence against civilians

Violence events in Mali

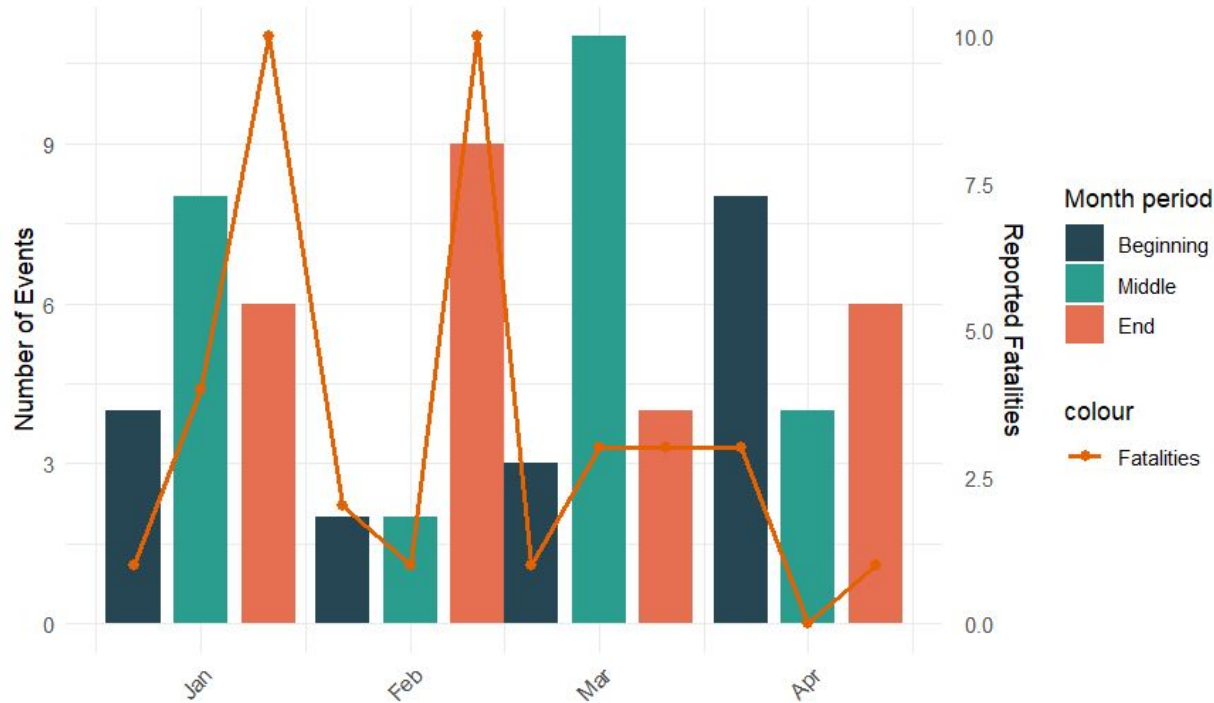
Violence events in Mali: 1 January 2025 – 28 April 2025



- The majority of events are concentrated in Segou and Mopti
- Most of the events with violence against civilians are also located here
- Violence against civilians also began to spread in Gao and Koulikoro, although there were isolated incidents before

Violence events and fatalities in Mali

Violence against civilians in Mali: 1 January 2025 – 28 April 2025

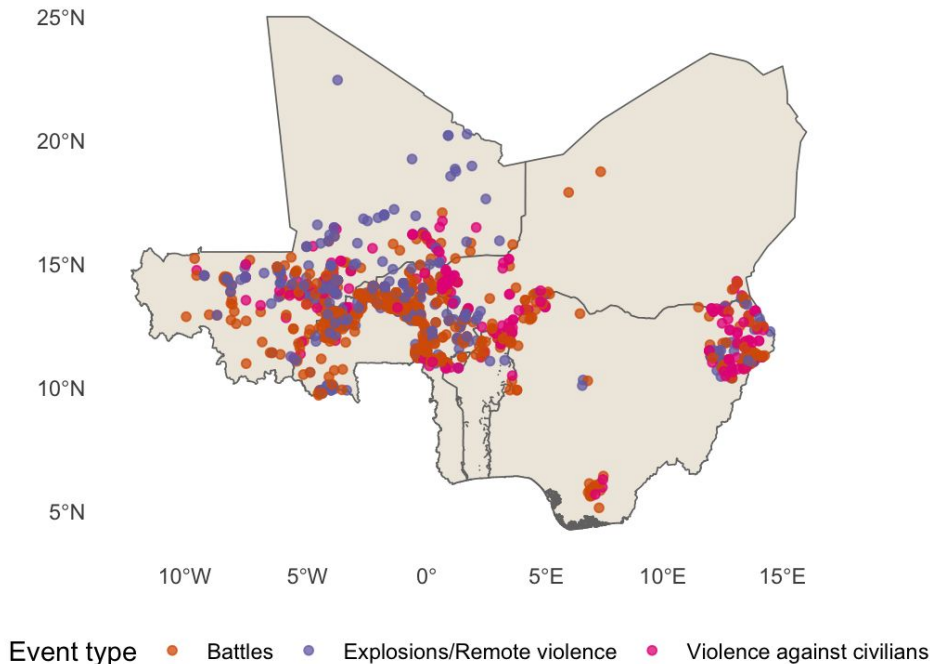


- The fatalities peaks occur at the ends of January and February
- The events peaks occur at the ends of February and middle of March
- The relationship between the number of events and fatalities is not strong

Expanding the Analysis: From Mali to the Region

Political violence events in the central Sahel and southern frontier

ACLED events, January–April 2025

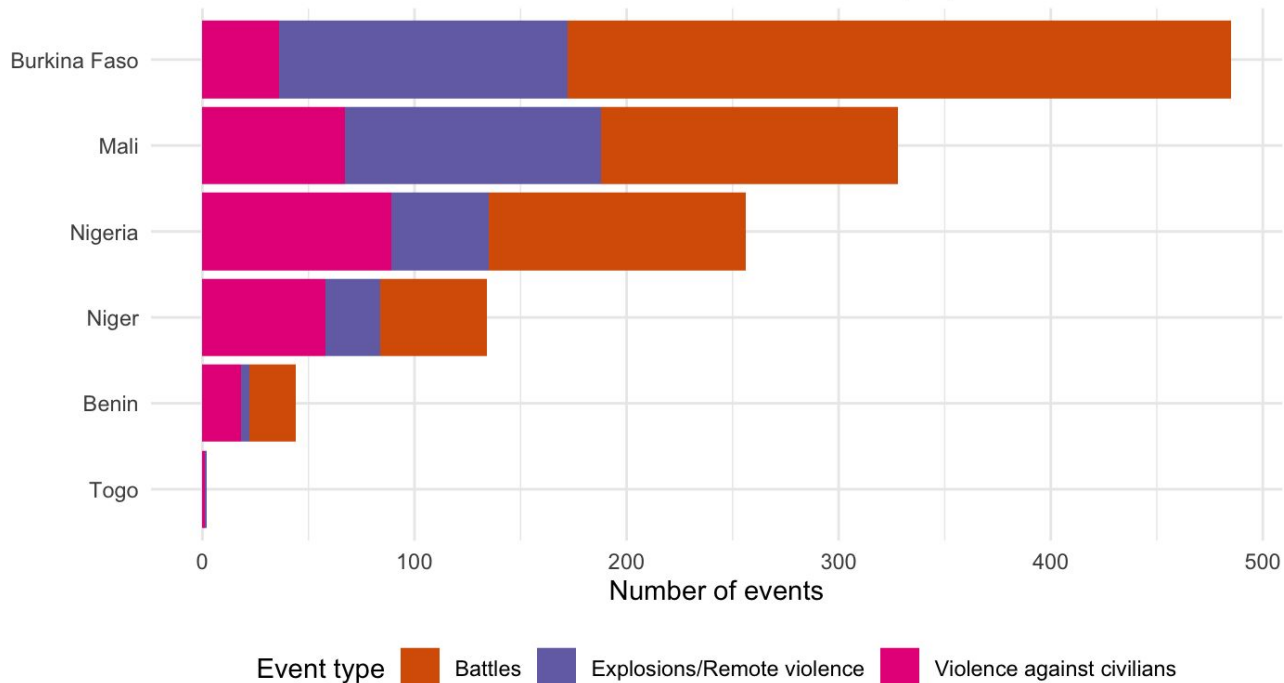


- **Conflict remains concentrated in the central Sahel**
Most events are located in Mali, Burkina Faso, and Niger.
- **Violence also appears along the southern frontier**
Events in Benin, Togo, and Nigeria show that insecurity is not fully limited to the central Sahel.

Political Violence Events by Country

Number of political violence events by country

Central Sahel and southern frontier, January–April 2025

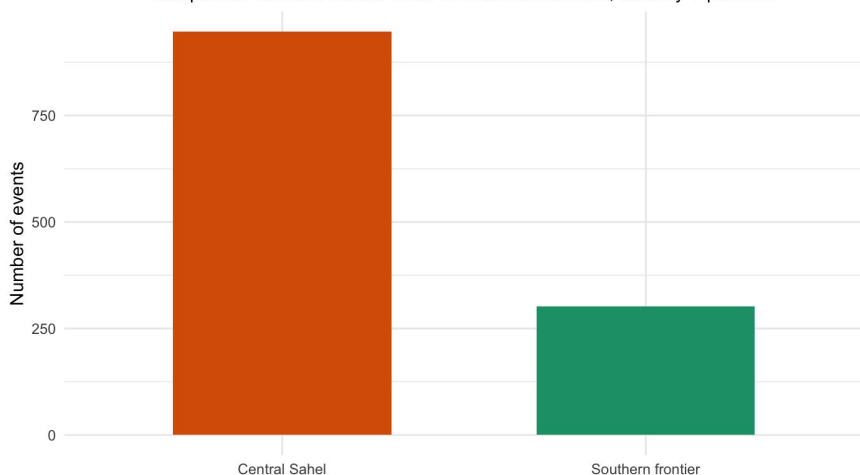


- **Burkina Faso records the highest number of events**
It is the main hotspot in our January–April 2025 dataset.
- **Mali remains a major conflict center**
Nigeria also has many events, but this may reflect both regional insecurity and Nigeria's own internal conflict dynamics.

Central Sahel vs Southern Frontier

Political violence remains concentrated in the central Sahel

Comparison between central Sahel and southern frontier, January–April 2025



The central Sahel has many more events

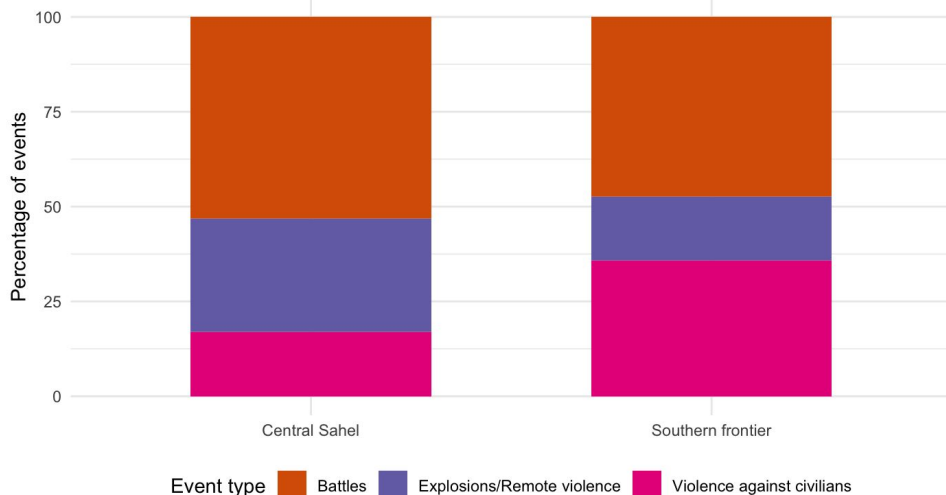
Mali, Burkina Faso, and Niger account for most political violence events.

The southern frontier is less intense but still affected

Benin, Togo, and Nigeria show lower event numbers, but they are not conflict-free.

Composition of political violence by regional group

Central Sahel compared with the southern frontier



The southern frontier has a higher share of violence against civilians

Even though it has fewer total events, civilian-targeting forms a larger percentage of violence there.

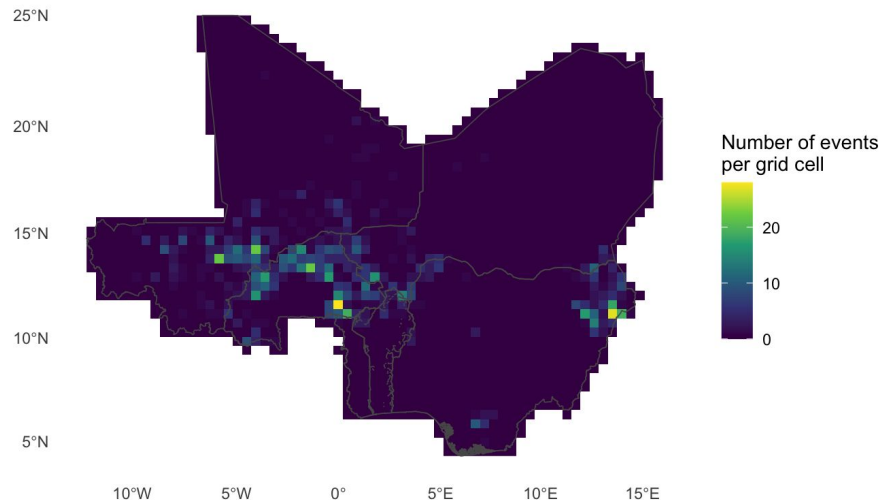
This suggests a different violence profile

The southern frontier is not only a weaker version of the central Sahel; the composition of violence changes by region.

Conflict Intensity Raster Map

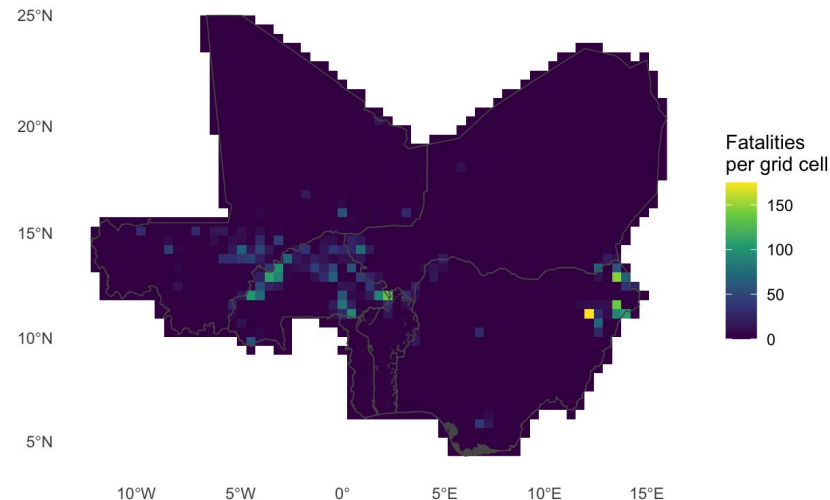
Rasterized conflict intensity in the central Sahel and southern frontier

ACLED political violence events aggregated into 50 km grid cells



Rasterized conflict severity in the central Sahel and southern frontier

Reported fatalities aggregated into 50 km grid cells



Raster analysis highlights the main hotspots

The highest conflict intensity appears in Burkina Faso, Mali, and nearby border areas.

Smaller hotspots appear farther south

This supports the idea of exposure along the southern frontier, especially around northern Benin and northeastern Nigeria.

Event frequency and severity are not the same

Some areas have many events, while others have fewer but deadlier events.

Fatalities are concentrated in specific hotspots

This shows why we analyze both number of events and reported fatalities.

Conflict correlation with natural resources



Location and correlation of mining areas and violences

- Mining sites and political violence do not initially appear correlated across the Sahara-Sahel belt.

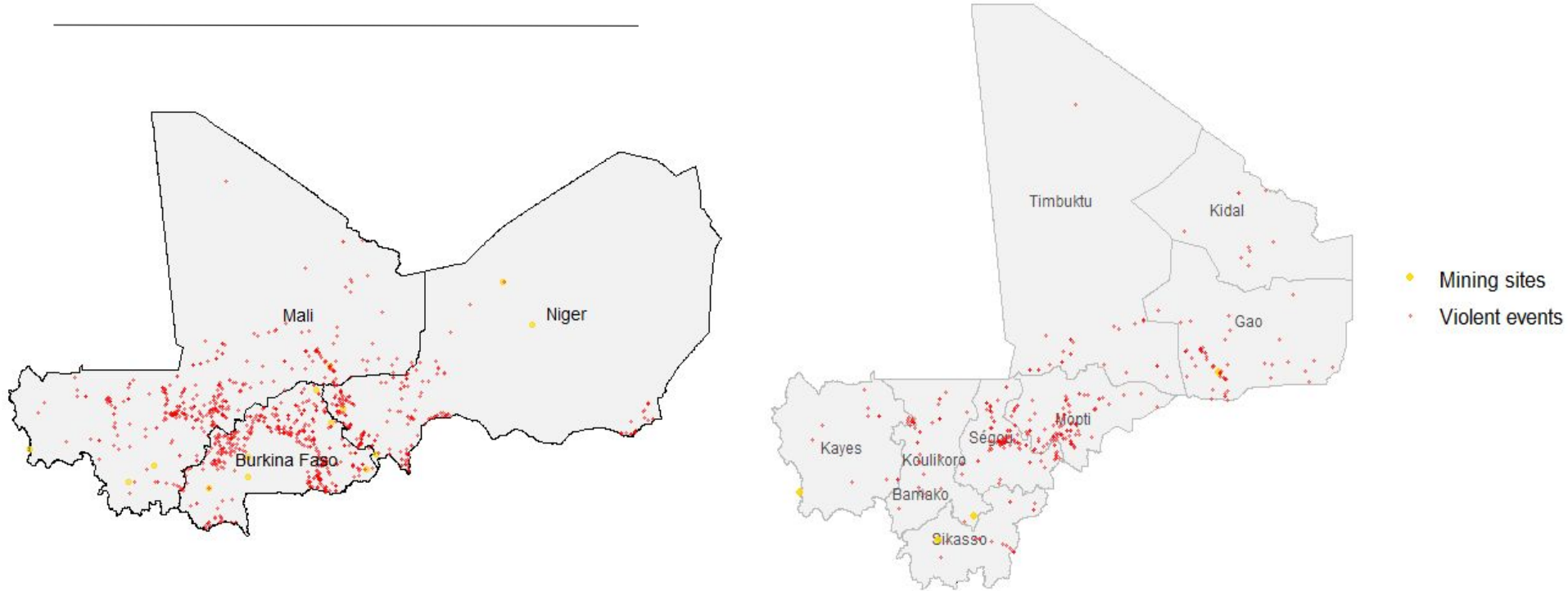
Narrowing the research on a specific area

- The research narrows to Mali and Burkina Faso to better identify spatial patterns.

Concentration of phenomena

- Mining zones and violent events seem concentrated in specific areas.

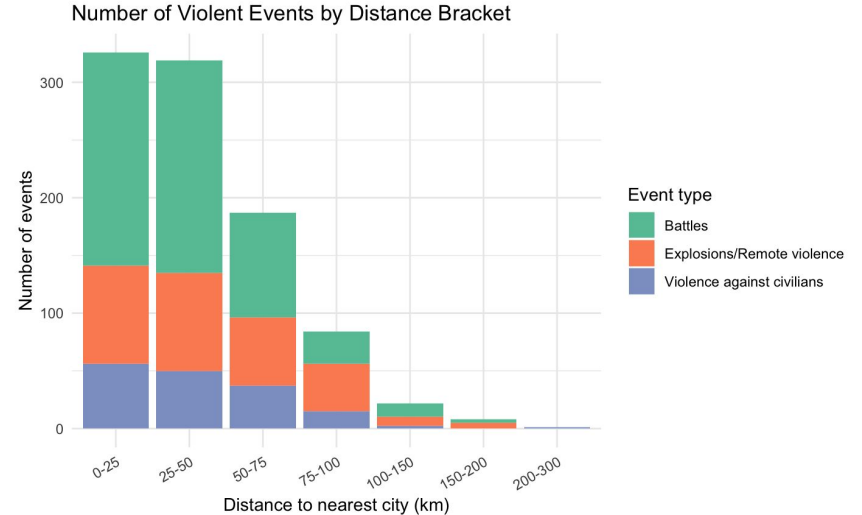
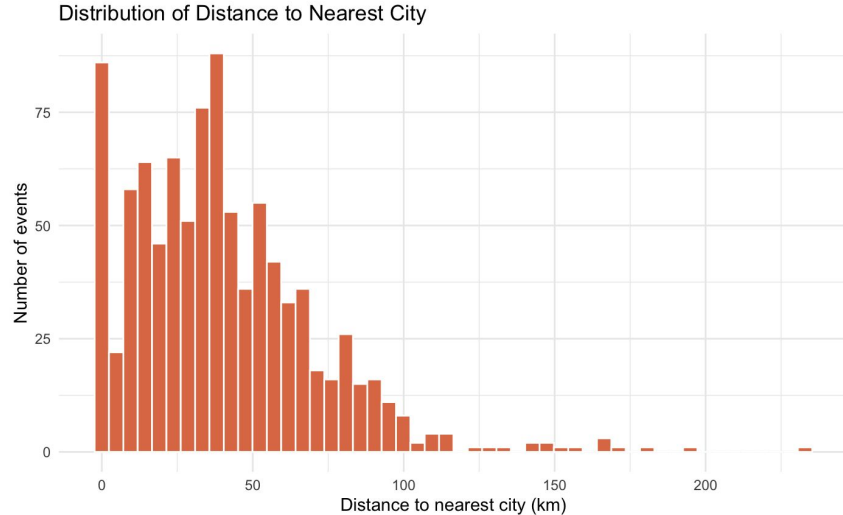
AES and Mali cases



After the narrowing, the analysis was focused on specific areas to highlight better the two phenomena repartition.

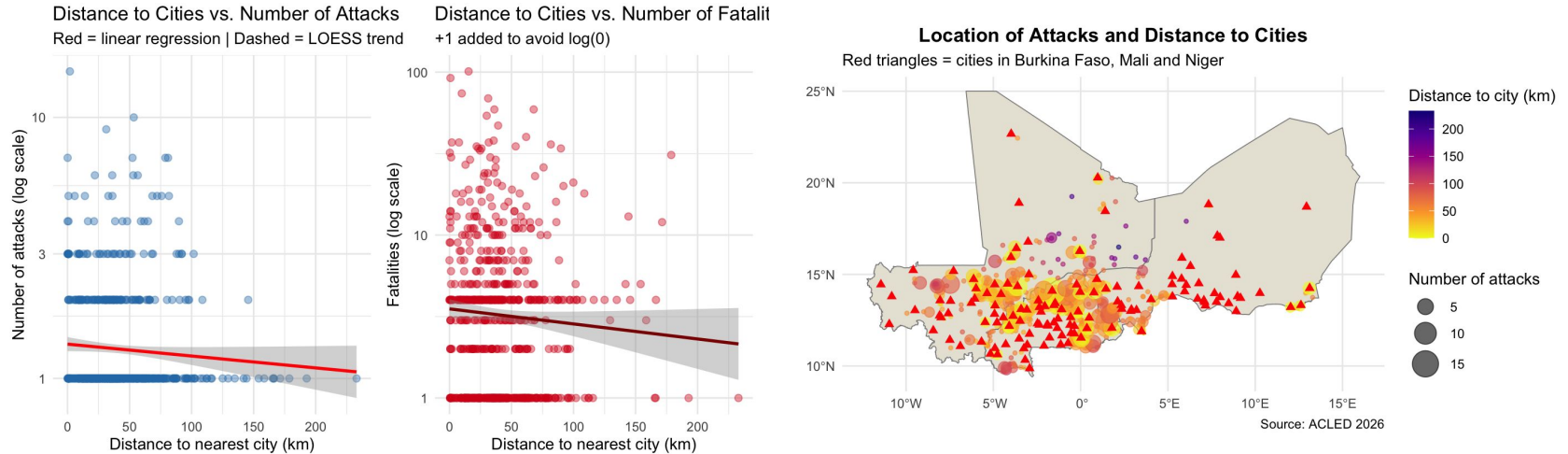
- The analysis was then focused on the AES states (Mali, Burkina Faso, Niger).
- A final focus on Mali aims to highlight possible correlations between mining sites and violence.

Distribution of terrorist attacks according to distance



- Violence is strongly concentrated near cities: the 0–25 km and 25–50 km brackets each account for ~325 events, by far the highest counts.
- A sharp drop-off occurs after 50 km - events fall from ~190 to ~85 between the 50-75 and 75-100 km brackets.
- Beyond 150 km, violence is nearly absent, suggesting a clear urban gravity effect.
- Battles dominate in all distance brackets; Explosions/Remote violence is consistently the second most common type.

Distribution of terrorist attacks according to distance



- Both regression lines (red) slope downward, confirming that attacks and fatalities are slightly more frequent closer to cities.
- The fatalities plot mirrors the same pattern: higher and more dispersed values cluster near 0 km, thinning out beyond 100 km.
- The wide confidence interval reflects a weak but real signal - distance has a measurable effect, but high variance means other factors are also at play.